# PORTFOLIO

CHARAN TEJA REDDY A M

May 19 2026

## General Problems in Mathematics

**1. Problem in a Plane mirror Field-of-view solved by using complex numbers.**

Consider a plane mirror of fixed length $a$. Suppose a point source $O$ is placed in front of the mirror as shown in fig 1.1. It is also given that the ends of the mirror make angle $\alpha$ at $O$ when subtended to meet. There is a straight screen which is placed to exactly fit the field of view of $O$ w.r.t. the mirror. Furthermore, the ends of the screen when subtended to meet at the virtual image have equal lengths. Show that:

$$\text{Screen length} \propto \left|\sin\frac{\alpha}{2}\right|$$

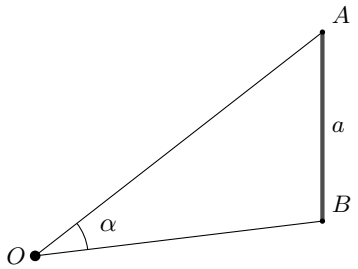



Figure 1.1: Geometric configuration of the mirror and the point source.

**2.**

We know that in the Binomial theorem for positive integral powers,

$$(1+x)^n = {}^nC_0 + {}^nC_1x + \cdots + {}^nC_nx^n, \qquad x \in \mathbb{R},\ n \in \mathbb{N}.$$

Similarly, let $W_n(x)$ be a function such that

$$W_n(x) = f^{(0)}(x) + {}^nC_1f^{(1)}(x) + {}^nC_2f^{(2)}(x) + \cdots + {}^nC_nf^{(n)}(x).$$

In a cleaner way,

$$\boxed{W_n(x) = {}^nC_0 f^{(0)}(x) + {}^nC_1 f^{(1)}(x) + {}^nC_2 f^{(2)}(x) + \cdots + {}^nC_n f^{(n)}(x)}$$

where $f(x)$ is an infinitely differentiable function over $x \in \mathbb{R}$ [i.e. differentiable of any order] and $f^{(r)}(x)$ denotes the $r$th derivative of $f(x)$; $f^{(0)}(x) = f(x)$.

For $W_n(x) = 0$, answer the following questions:

(i) Solve for $f(x)$.

(ii) What can be the degree of $f(x)$, if applicable?

(iii) How many roots can $f(x)$ have at most, and at least?

(iv) Suppose $S_i(x)$ is a function which is differentiable over $x \in \mathbb{R}$. Find the number of such $S_i(x)$ for which

$$\int_a^b f(x)\,\mathrm{d}x + \int_{f(a)}^{f(b)} S_i(x)\,\mathrm{d}x = b\,f(b) - a\,f(a); \qquad a, b \in \mathbb{R},\ a \neq b.$$

Assume that $f(x)$ is fixed w.r.t. $S_i(x)$.

**3. Basic**

Consider the following statements. For two differentiable functions $f, g : \mathbb{R} \to \mathbb{R}$,

$S_1$: There exists a $c$ such that $c \in (a, b)$ and

$$\frac{g'(c)}{f'(c)} = \frac{g(b) - g(a)}{f(b) - f(a)}; \qquad f(b) \neq f(a).$$

$S_2$: There exists a $c \in (a, b)$ such that $g'(c) = \dfrac{g(a) - g(b)}{a - b}$.

$S_3$: There exists a $c \in (a, b)$ such that $f'(c) = \dfrac{f(a) - f(b)}{a - b}$.

$S_4$: If for $x \in [a, b]$, $f(a) = f(b)$ or $g(a) = g(b)$, then there exists a $c \in (a, b)$ such that $f'(c) = 0$ or $g'(c) = 0$ respectively.

[In all the above statements, $a, b, c \in \mathbb{R}$.]
Which of the following is/are correct?

(A) $S_1 \Rightarrow S_2$ or $S_1 \Rightarrow S_3$

(B) '$S_1$ and $S_2$' $\Rightarrow S_4$

(C) $S_2$ and $S_3 \Rightarrow S_1$

(D) '$S_4$' is derived from $S_1, S_2, S_3$ (any one of them).

**4.**

Let $I(x) = \int \frac{a(x)}{\big(b(x)\big)^2}\,\mathrm{d}x$, where $a(x)$ and $b(x)$ are real valued functions, $b(x) \neq 0$. Show that there exists a function $f(x)$, which is real valued, such that

$$\frac{f(x)}{I(x)} = b(x); \qquad I(x) \neq 0,$$

if not, then disprove it.

**5.** Consider a function $f(x) = {}^{n}C_{x}$ defined for $n \neq 0$, $0 \leq x \leq n$; $x, n \in \mathbb{Z} - \mathbb{Z}^{-}$ where $\boxed{{}^{n}C_{r} = \frac{n!}{(n-r)!r!}}$. In the $XY$-plane, if $f(x)$ is plotted on it then find the area under the path traced by connecting its points with a single straight lines (that is, connecting every two consecutive points with a single straight line).

**6.** Consider a function $f(x)$ such that

$$\sum_{n=1}^{k} f^{(4n)}(x) = 0,$$

where $f^{(r)}(x)$ denotes the $r$-th derivative of $f(x)$ and $k \in \mathbb{N}$. Then which of the following is/are true?

(A) All such $f(x)$ obeying the above equation are periodic.

(B) Infinitely many such $f(x)$ exist that satisfy the equation.

(C) The number of such $f(x)$ satisfying the equation is finite.

(D) There exist infinitely many $f(x)$ which are periodic and satisfy this equation.

**7. Counting NENO combinations:**

Let $f(x), g(x), h(x)$ be three real-valued functions, none of which is NENO (neither even nor odd) — that is, each of $f, g, h$ is either even or odd. Using each of $f, g, h$ at most once (no repetition), and combining them only with basic arithmetic operations and composition of functions, find the number of distinct functions $\alpha(x)$ that can be formed such that $\alpha(x)$ is NENO (neither even nor odd). Take $f, g, h$ to be fixed with respect to $\alpha(x)$.

**8.** If $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function such that

$$f(x) - 4f\left(\frac{x}{3}\right) + 2f\left(\frac{x}{a^2}\right) = 3x^2 \quad \forall x \in \mathbb{R},\ (a \in \mathbb{R},\ a \neq 0),$$

then:

(i) What is the nature of $f$ (sinusoidal, exponential, polynomial, etc.)?

(ii) Prove the nature of $f$ from the answer you got in (i).

(iii) If $f(\sqrt{8}) = 36$, then find all possible values of $a^4$.

(iv) Find all such possible $f(x)$.

**9.** Evaluate the integral (on any interval where $e^x + \sin x \neq 0$):

$$\int \frac{1 + 2e^x \sin x}{(e^x + \sin x)^2}\, dx.$$

**10.** Suppose the graph of $K \sin x$ ($K \in \mathbb{R}^+$) is taken over $n$ periods, i.e. over an interval of length $2\pi n$, and is cut at its ends and joined to form a circle $C_0$ of circumference $2\pi n$ (hence of radius $n$), with the $K \sin x$ curve drawn along it so that a closed, wavefront-type figure is formed. The peak points of the wave point towards the centre of $C_0$.

Now a circle $C_1$ is created internally by joining the peak points of $K \sin x$, and the graph of $\frac{K}{2} \sin x$ (over the same $n$ periods) is drawn on $C_1$ in the same manner. This process goes on: for every $j \geq 0$, the circle $C_{j+1}$ passes through the peak points of the wave $\frac{K}{2^j} \sin x$ drawn on $C_j$. Assume the process continues indefinitely (every $C_j$ has a positive radius).

Let $A$ be the sum of the areas of all the circles $C_0, C_1, C_2, \ldots$ Then answer the following:

(i) Find $A$. For what relation between $n$ and $K$ is $A$ finite?

(ii) On what parameters does $A$ vary (when it is finite)?

(iii) Let $B$ be the sum of the areas of the ring-shaped regions between $C_j$ and $C_{j+1}$ for $j = 0, 1, 2, \ldots$ Find $B$ in terms of $n$ and $K$.